

G - Count Holidays

分值: 600 分

Problem Statement

Takahashi is creating a work schedule for N days by designating each day as a workday or a holiday.

高桥正在制定 N 天的工作安排，他需要将每一天设定为工作日或休息日。

You are given a string S of length N representing the constraints on work days. If the i -th character of S is x , day i must be a workday. If it is $.$, day i can be either a workday or a holiday.

给定一个长度为 N 的字符串 S ，它表示工作日的约束条件。如果 S 的第 i 个字符是 x ，那么第 i 天必须是工作日；如果是 $.$ ，那么第 i 天既可以是工作日也可以是休息日。

There are 2^q valid work schedules satisfying the constraints, where q is the number of $.$ characters in S . For each $k = 1, 2, \dots, N$, solve the following problem:

满足约束条件的有效工作安排共有 2^q 种，其中 q 是 S 中 $.$ 字符的个数。对于每个 $k = 1, 2, \dots, N$ ，求解以下问题：

Among the 2^q valid work schedules satisfying the constraints, find the number, modulo 998244353, of schedules in which the longest consecutive block of holidays is exactly k days.

在所有满足约束条件的 2^q 种有效工作安排中，统计最长连续休息日恰好为 k 天的安排数量，结果对 998244353 取模。

Constraints

- N is an integer between 1 and 2×10^5 , inclusive.
- S is a string of length N consisting of $.$, x .
- N 是介于 1 和 2×10^5 之间（含两端）的整数。
- S 是长度为 N 的字符串，仅由 $.$ 和 x 组成。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

N
 S

Output

Output N lines. The i -th line should contain the answer for $k = i$.

输出 N 行，第 i 行应包含 $k = i$ 对应的答案。

Sample Input 1

```
5
.X...
```

Sample Output 1

```
9
4
2
0
0
```

Denoting holidays as o, the valid work schedules are as follows:

用 o 表示休息日，有效工作安排如下：

- $k = 1$: oxxxx, oxoxx, oxoxo, oxxox, oxxxo, xxoxx, xxoxo, xxxox, xxxxo
- $k = 2$: oxoox, oxoo, xxoox, xxxoo
- $k = 3$: oxooo, xxooo
- $k = 1$: oxxxx, oxoxx, oxoxo, oxxox, oxxxo, xxoxx, xxoxo, xxxox, xxxxo

- $k = 2$: 0x00x, 0xx00, xx00x, xxx00
- $k = 3$: 0x000, xx000

Sample Input 2

```
7
.....
```

Sample Output 2

```
33
47
27
12
5
2
1
```

Sample Input 3

```
20
.....X...X.....
```

Sample Output 3

```
9359
75312
94664
46840
23680
7168
3072
1280
512
256
```

0
0
0
0
0
0
0
0
0
0
0

